SOFTWARE REQUIREMENTS SPECIFICATION

Execution and Traceability Update — Snake Game

Supplement to SRS Version 1.0 (Draft for Review), dated 12 October 2026

Project	Snake Game — Software Engineering Mini-Project
Prepared by	Project team: Charan, Vrunda, Ved and Tushar
Purpose	Record implementation-aligned environment and verification status
Automated suite	68 tests passed; 0 failed
Evidence	reports/Automated_Test_Results_68_Tests.png
Document status	Supplement only; original SRS approval fields remain pending

Document control note

The source SRS remains Version 1.0 and identifies itself as Draft for Review. This addendum records test execution information without changing the original requirement wording, use-case model or approval status. The source SRS places the use-case diagram on page 6 and the RTM across pages 6–7.

1. Implementation and operating environment update

The implementation details below are based on the project's README and CMake configuration supplied for review. The project uses C++17, CMake and SFML 2.6.x; the README documents the keyboard controls and build/test workflow.

Item	Recorded value
Application type	Single-player local desktop game
Language standard	C++17
Build system	CMake 4.3.1
Compiler	Microsoft C/C++ compiler 19.52.36815 for x64 (Preview)
Operating system	Windows 11 Home Single Language, build 26300, 64-bit
Hardware	HP Victus 15-fa1xxx; Intel Core i5-13420H; 16 GB RAM; NVIDIA RTX 4050 Laptop GPU and Intel UHD Graphics
Graphics library	SFML 2.6.2
Controls	Enter/Space starts; arrow keys steer; R/r restarts after Game Over/Win; Escape exits
Score rule	10 points per food; best score for the session is retained across restart
Initial/restart state	Snake resets to 3 segments at (5,5), direction Right; current score resets to 0; best score is retained

2. Verification summary

The captured CTest output reports 100% tests passed, 0 failed out of 68, with a total real test time of 25.14 seconds. The test run was executed in PowerShell after adding the SFML 2.6.2 runtime directory to PATH. This automated result does not, by itself, replace the separate manual and security procedures in the Test Plan.

Test reference	SRS requirements	Status	Basis / qualification
TC-01	FR-01–FR-03	Pass (reported)	Startup and initial display checks reported complete.
TC-02	FR-04–FR-06	Pass (reported)	Movement and reversal checks reported complete; automated movement tests passed.
TC-03	FR-07–FR-09	Pass (reported)	Food placement, detection and growth checks reported complete; automated food tests passed.
TC-04	FR-10–FR-11	Pass (reported)	Score and replacement-food checks reported complete; automated score tests passed.
TC-05	FR-12–FR-15	Pass (reported)	Boundary/self-collision and departing-tail checks reported complete; automated collision/integration tests passed.
TC-06	FR-16–FR-17	Pass (reported)	Restart/reset checks reported complete; automated restart tests passed.
TC-07	FR-18, SR-05	Manual pass reported	Sanitizer-enabled sub-steps from the Test Plan have not been confirmed.
TC-08	NFR-01	Pass (reported)	Three 60-second runs reportedly averaged approximately 60 FPS each (target ≥30 FPS).
TC-09	NFR-02	Pass (reported)	External reviewer reportedly understood controls without assistance.
TC-10	NFR-03	Pass (reported)	Ten consecutive start/game-over/restart cycles reportedly completed without crash/freeze.
TC-11	NFR-04	Pass (reported)	Source modules reportedly reviewed and responsibilities identified.
TC-12	NFR-05, NFR-06	Pass (reported)	Teammate reportedly completed fresh clone, build and launch.
TC-13	SR-01, SR-04	Manual pass reported	Input and reversal checks reported successful; sanitizer sub-step not confirmed.
TC-14	SR-02, SR-03	Partial	Static source searches reportedly found no command-execution or unrelated-file-access API matches. Full Process Monitor trace and complete review not confirmed.

3. Updated acceptance summary

The automated suite is fully passing at 68/68. Reported manual checks support the functional and non-functional outcomes above. However, overall security acceptance remains open: TC-14 is partial, and sanitizer execution for TC-07 and TC-13 has not been confirmed. The SRS should therefore not state that every requirement has been fully accepted yet.

4. RTM status guidance for SRS Section 8

Use the following status convention in the SRS RTM and keep it synchronized with Section 10 of the Test Plan and the project RTM workbook:

Requirements	Suggested status	Reason
FR-01 to FR-18	P (Pass) — based on reported TC-01 to TC-07 results	Functional cases were reported as passed; automated tests also support relevant logic.
NFR-01 to NFR-06	P (Pass) — based on reported TC-08 to TC-12 results	Performance, usability, reliability, maintainability and fresh-clone checks were reported passed.
SR-01 and SR-04	P (Pass) — manual verification only	TC-13 manual input checks reported passed; sanitizer sub-step remains unconfirmed.
SR-05	P (Pass) — manual verification only	TC-07 manual safe-close check reported passed; sanitizer sub-step remains unconfirmed.
SR-02 and SR-03	N / Partial (do not mark P yet)	TC-14 full review and runtime Process Monitor evidence are still outstanding.

5. Outstanding actions before final approval

1. Complete TC-14 code review of every file-open/read/write call and capture a Windows Process Monitor trace through a full round, restart and application close.

2. Run or formally record the sanitizer-enabled verification required by TC-07 and TC-13.

3. Synchronize the SRS Section 8 RTM, Test Plan Section 10 RTM and RTM_SnakeGame.xlsx. Do not mark SR-02/SR-03 fully passed until TC-14 is complete.

4. Update the original SRS revision history after the team approves this supplement. Do not fill instructor/team signatures or approval dates without the actual approvals.

5. Retain the CTest screenshot at reports/Automated_Test_Results_68_Tests.png and attach additional manual/security evidence to the repository.

6. Conclusion

The Snake Game automated test suite completed with 68 of 68 tests passing. Reported manual checks also indicate successful functional, performance, usability, reliability, maintainability and compatibility verification. The SRS remains a draft until the outstanding security verification and formal document approvals are completed. This addendum records the current evidence and should be stored alongside, not instead of, the original SRS.

End of SRS execution update